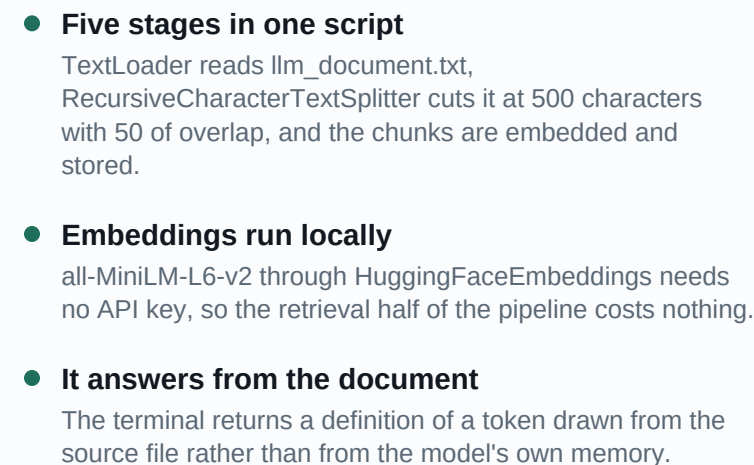
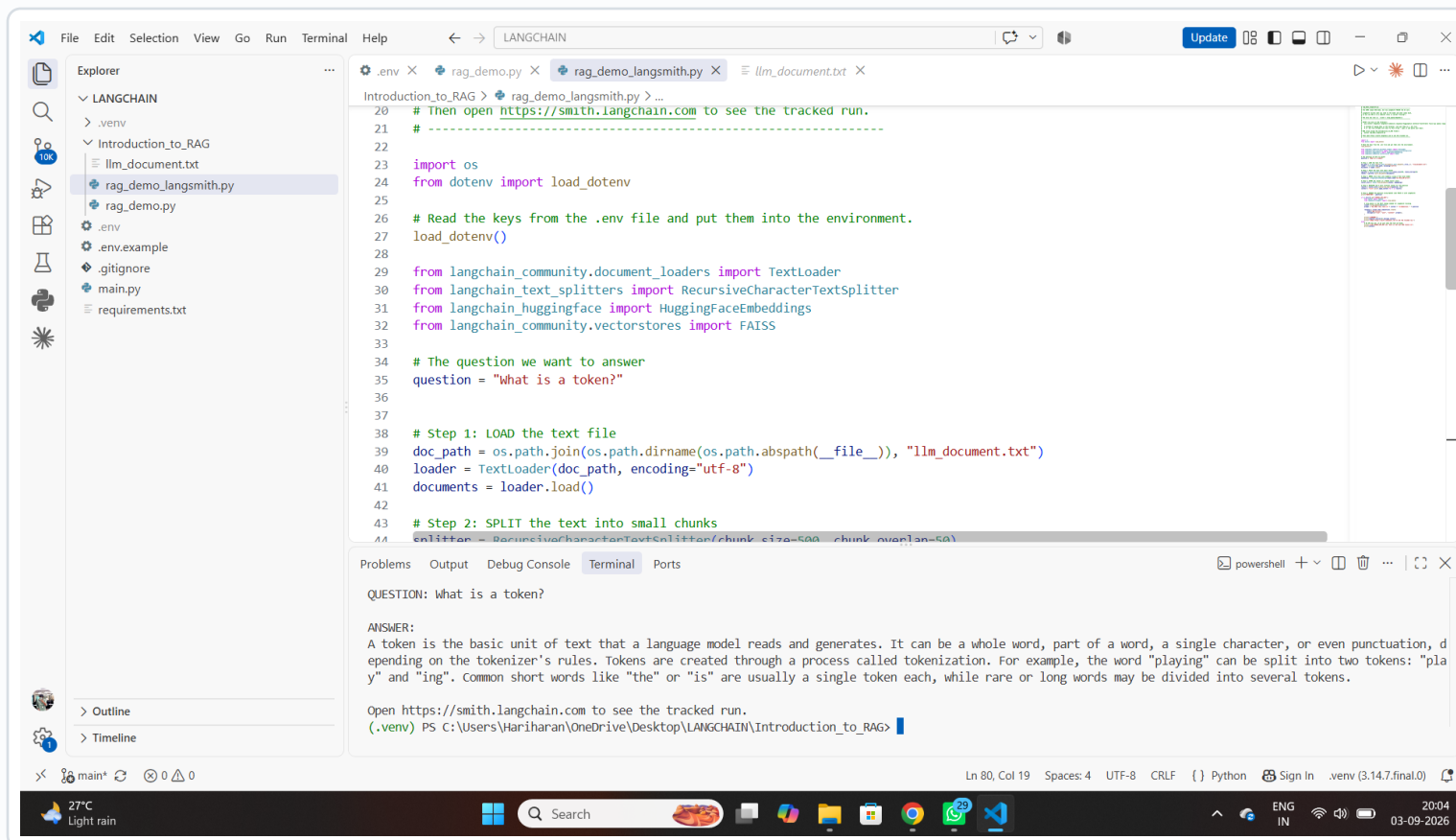


Building the RAG Pipeline



STEP 02

Adding LangSmith Tracing



```
File Edit Selection View Go Run Terminal Help
LANGCHAIN
Update
.rag_demo.py X rag_demo_langsmith.py X llm_document.txt X
Introduction_to_RAG > rag_demo_langsmith.py > ...
20 # Then open https://smith.langchain.com to see the tracked run.
21 # -----
22
23 import os
24 from dotenv import load_dotenv
25
26 # Read the keys from the .env file and put them into the environment.
27 load_dotenv()
28
29 from langchain_community.document_loaders import TextLoader
30 from langchain_text_splitters import RecursiveCharacterTextSplitter
31 from langchain_huggingface import HuggingFaceEmbeddings
32 from langchain_community.vectorstores import FAISS
33
34 # The question we want to answer
35 question = "What is a token?"
36
37
38 # Step 1: LOAD the text file
39 doc_path = os.path.join(os.path.dirname(os.path.abspath(__file__)), "llm_document.txt")
40 loader = TextLoader(doc_path, encoding="utf-8")
41 documents = loader.load()
42
43 # Step 2: SPLIT the text into small chunks
44 splitter = RecursiveCharacterTextSplitter(chunk_size=500, chunk_overlap=50)
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STEP 03

Inspecting the Traced Run

The screenshot displays the LangSmith application interface, which is used for monitoring and managing LLM applications. The interface is organized into three main sections:

- Left Sidebar (Navigation):** Contains various application management tools such as "All applications", "Search", "Home", "Tracing" (selected), "Engine", "Monitoring", "Datasets & Experiments", "Evaluators", "Annotation Queues", "Custom Apps", "Prompts", "Playground", "Studio", "Context Hub", "Deployments", "Sandboxes", "LLM Gateway", and "Keyboard Shortcuts".
- Central Panel (Tracing List):** Shows a list of traces for the "langchain-project". The selected trace is "ChatOpenAI" with a status of "Success" and a cost of "\$0.0001".
- Right Panel (Trace Details):** Provides a detailed view of the selected trace, including the "Feedback" section (with a "Give Feedback" button) and the "Input" section. The input text is:

Use ONLY this text:
What is a token?
A token is the basic unit of text that a language model reads and generates. Before a model can process text, the text must be broken down into tokens through a process called tokenization. A token can be a whole word, part of a word, a single character, or even punctuation, depending on the tokenizer's rules. For example, the word "playing" might be split into two tokens: "play" and "ing". Common short words like "the" or "is" are often a single token each, while rare or long words may be split into several tokens.
Why tokens matter
Language models do not understand raw text directly. Instead, each token is converted into a numerical representation (an embedding) that the model can process mathematically. The number of tokens in a piece of text also determines how much it costs to process with an API, and it affects the maximum amount of text (the context window) that a model can read or generate in a single request.
Question: What is a token?

The bottom of the image shows a Windows taskbar with the date and time as 20:04 on 03-09-2026, and a system tray with various icons including weather, search, and network status.

- **The full prompt is visible**
The trace shows the retrieved passage passed in under 'Use ONLY this text', which confirms retrieval worked before the model was asked anything.
- **341 tokens, not 118**
The count is roughly triple an ordinary call, because the retrieved context travels with the question. This is what RAG costs.
- **Measured, not guessed**
gpt-4o-mini at 2.74s and under \$0.0001, with input, output and metadata on one screen for every run.